DATA 211 – Introduction to Data Science

PRE-PROCESSING

Prof Dr Usman Qamar

Data Acquisition

- *Acquisition: act of acquiring something.*
- *Data Acquisition: acquiring data for analysis.*

It could be done via:

- **Direct approach:** is a process that converts analog and/or real world data into digital data that can be processed for further analysis. Data Acquisition System (DAS) are used in this acquisition.
- **Indirect approach:** is a process of identifying sources/vendors that provides the data based on specifications.

Data Collections

- Collections of data available for analysis. For example: Open data repositories and Social media data.
- Legality and usage rights should be considered before using data from collections.

Data Storage

- Structured text data (*the data we consider in this course*) can be stored in the following formats:
  - Comma Separated Values **CSV**
  - Tab Separated Values **TSV**
  - eXtensible Markup Language **XML**
  - Really Simple Syndication **RSS**
  - JavaScript Object Notation **JSON**
- We will be mostly working with the CSV or TSV files in this course. However, data in any of the above forms can be read into Python (or any programming language of your choice) using standard procedures.

Why Data Preprocessing?

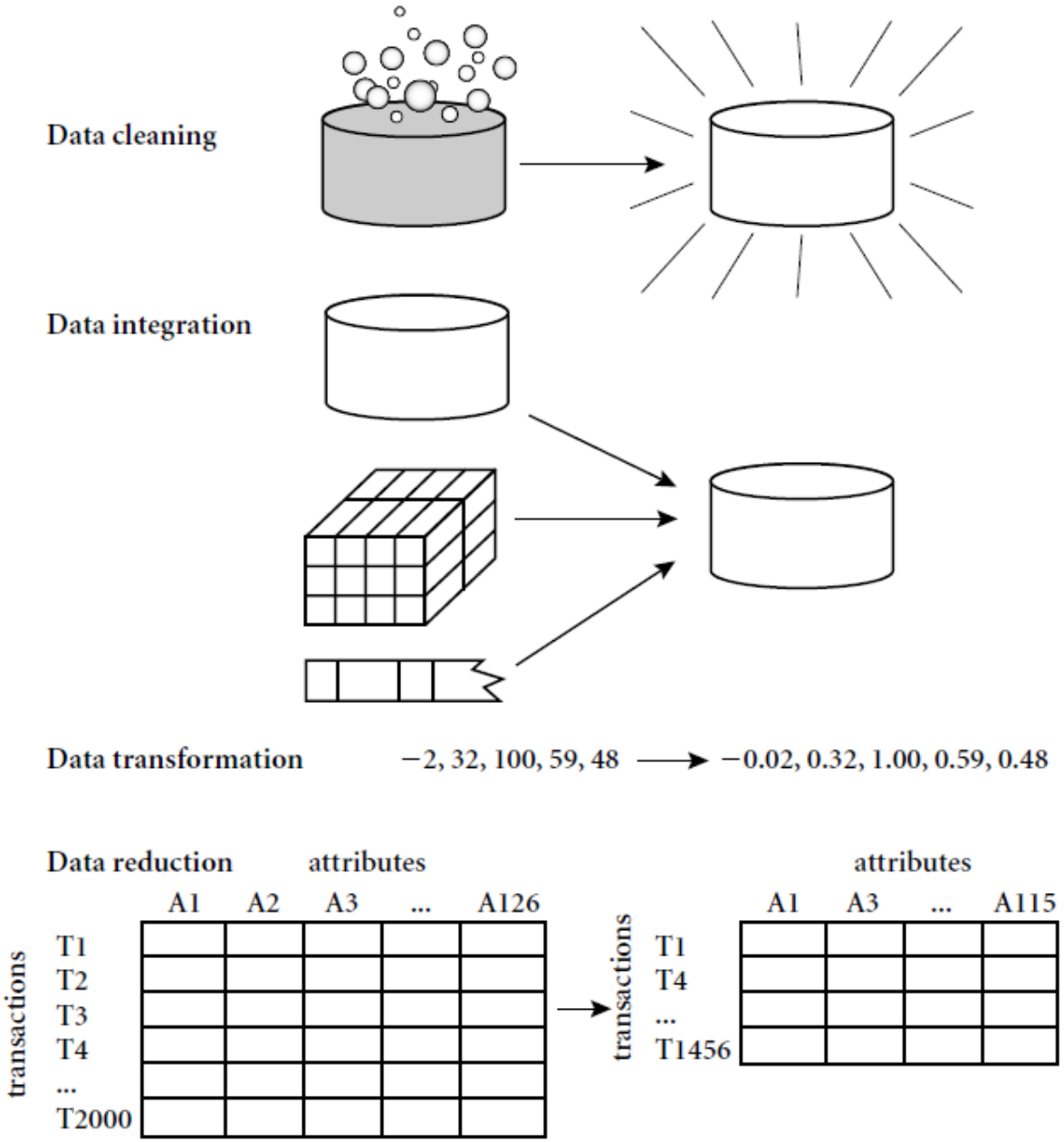
- Data in the real world is dirty
  - incomplete: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
    - e.g., occupation=“ ”
  - noisy: containing errors or outliers
    - e.g., Salary=“-10”
  - inconsistent: containing discrepancies in codes or names
    - e.g., Age=“42” Birthday=“03/07/1997”
    - e.g., Was rating “1,2,3”, now rating “A, B, C”
    - e.g., discrepancy between duplicate records

Why Is Data Preprocessing Important?

- No quality data, no quality mining results!
  - Quality decisions must be based on quality data
    - e.g., duplicate or missing data may cause incorrect or even misleading statistics
  - Data warehouse needs consistent integration of quality data
- Data extraction, cleaning, and transformation comprises the majority of the work of building a data warehouse

Major Tasks in Data Preprocessing

- Data cleaning
  - Fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies
- Data integration
  - Integration of multiple databases, data cubes, or files
- Data transformation
  - Normalization and aggregation
- Data reduction
  - Obtains reduced representation in volume but produces the same or similar analytical results
- Data discretization
  - Part of data reduction but with particular importance, especially for numerical data



Forms of data preprocessing.

How to Handle Missing Data?

- Ignore the tuple:.
- Fill in the missing value manually: tedious + infeasible?
- Fill in automatically with
  - a global constant : e.g., “unknown”, a new class?!
  - the attribute mean, averger or mode

Data Transformation

- Normalization: scaled to fall within a small, specified range
  - min-max normalization
  - normalization by decimal scaling
- Attribute/feature construction
  - New attributes constructed from the given ones
    - Area computation from height and width
    - BMI computation from weight and height

Standard Scaler

Scores	Scaled Scores
1	-1.714816
2	-1.680173
3	-1.645531
⋮	⋮
98	1.645531
99	1.680173
100	1.714816

μ	50.5	0
σ	28.86	1

Label Encoder

Scores	Encoded Scores
banana	1
apple	0
mango	2
banana	1
orange	3
banana	1
orange	3

Min-Max Scaler

Scores	Scaled Scores
1	0
2	0.0101
3	0.0202
⋮	⋮
98	0.9797
99	0.9898
100	1

One Hot Encoding

Color	Color_Red	Color_Green	Color_Blue	Color_Black
Red	1	0	0	0
Green	0	1	0	0
Blue	0	0	1	0
Black	0	0	0	1

Data Transformation: Normalization

Min-max normalization: to [new_min$_A$, new_max$_A$]

$$v' = \frac{v - min_A}{max_A - min_A}(new_max_A - new_min_A) + new_min_A$$

Ex. Let income range \$12,000 to \$98,000 normalized to [0.0, 1.0]. Then \$73,600 is mapped to $\frac{73{,}600 - 12{,}000}{98{,}000 - 12{,}000}(1.0 - 0) + 0 = 0.716$

End of lecture